

Transforming Production Planning Through Data Intelligence

Summary document

Egemen Alkan
Student Bachelor Applied Computer Science

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1. Introduction

The weekly planning meetings at Estée Lauder, typically held every Thursday, are critical for operational success but have historically been reactive and time-consuming. Planners currently spend a significant portion of their limited time manually validating production plans against 10 standard planning rules. This manual validation process forces the meetings to focus on data verification rather than strategic decision-making.

This internship project was designed to address this operational bottleneck through the realization of two complementary deliverables. The primary deliverable, an Enhanced Weekly Planning Report, transforms the existing Power BI environment into an automated violation tracker, creating a single source of truth that automatically identifies deviations from planning standards. The secondary deliverable, an AI Planning Assistant, consists of a formal feasibility study and prototype designed to help planners query data and procedural documentation using natural language. This document summarizes the technical realization, challenges, and value generated by these solutions.

2. Analysis

The project required a distinct analytical approach for each deliverable: a technical deep dive into existing tools for the primary reporting deliverable, and a comparative market analysis for the secondary AI deliverable.

2.1. Tools for the Primary Deliverable (Power BI Report)

The Power BI platform was pre-selected for this project, so the analysis focused on leveraging its advanced capabilities to model complex business logic.

- **Power BI Desktop & DAX:** Used for data modelling and creating the calculation engine. Simple aggregations were insufficient; complex DAX logic was required to translate abstract planning rules into concrete flags, particularly for complex scenarios like setup group changeovers and duplicate checks .
- **Power Apps:** Selected to integrate a write-back capability directly into the report, allowing planners to interact with the data and log decisions without leaving the dashboard.

2.2. Tools for the Secondary Deliverable (AI Assistant Research)

The AI workstream was conducted as a feasibility study to de-risk the company's investment in new technology. A comparative analysis was performed on three potential platforms against non-negotiable enterprise requirements: Security (Entra ID), Data Governance (in-tenant processing), and Live Data Connectivity.

- **Microsoft Copilot Studio:** Analysed as the enterprise-grade, low-code solution.
- **OpenAI Agent Builder:** Analysed for its ease of use versus its security compliance.
- **Custom Python/React:** Analysed for maximum customizability versus high development cost.

3. Realisation

3.1. Primary Deliverable: Enhanced Weekly Planning Report

The core objective of the enhanced report was to transform the weekly meeting from a data validation session into a strategic decision-making forum. The final solution acts as an automated violation tracker, creating a single source of truth that identifies deviations from standard planning rules.

3.1.1. Key Features & Structure

The report is structured to guide users from a high-level summary to actionable details:

- **Meeting Planning Dashboard:** A "triage" interface that summarizes violations per week and production line, allowing planners to quickly scan for risk.
- **Specialized Analysis Pages:** Dedicated views for *Demand & Attainment*, *PV1 Adherence*, and *Setup Group Efficiency*.
- **Order Compliance:** A granular to-do list identifying specific violations like Minimum Order Quantity (MOQ) or duplicates.

3.1.2. Technical Implementation Highlights

The transformation of abstract business rules into automated logic required complex technical solutions:

- **Context-Aware Changeovers (Rule 8):** A standard count was insufficient because the definition of a "changeover" varies by line type. Complex DAX logic was implemented to compare character strings (first 5 vs. first 8 characters) based on the specific resource type (Tubes vs. Vials) .
- **Two-Step Duplicate Validation (Rule 10):** To avoid false positives, the logic was refined to first check if an order exceeds the Maximum Order Quantity (MAXQ). Only small orders trigger a secondary check for other orders of the same material within a 28-day window .

3.1.3. Interactive Workflow: The Power App

To solve the issue of repeated flagging, a Power App was embedded directly into the report. This allows planners to "Mark as Agreed" specific violations, writing the decision to a central log and removing the flag from future meetings.

3.2. Secondary Deliverable: AI Planning Assistant

This workstream delivered a formal feasibility study and a high-fidelity prototype, validating the potential for an AI "Co-Pilot" to assist with data queries and procedural guidance.

3.2.1. Feasibility Study Findings

The research concluded that **Microsoft Copilot Studio** is the only viable platform for the company.

- **OpenAI Agent Builder** was deemed non-compliant because it cannot use Entra ID for authentication and processes data on external servers, violating data governance policies.
- **Microsoft Copilot Studio** met all requirements: native Entra ID authentication, in-tenant data processing, and live connectivity to SharePoint and Power BI .

3.2.2. Prototype Architecture & Success

A production-ready prototype was built in Copilot Studio to validate the recommendation.

- **Multi-Agent Routing:** The system uses a "Main Agent" to classify user intent and route questions to specialized child agents (e.g., a "Missing Parts Agent", "Ready To Firm Agent", or "Training Knowledge Topic").
- **Solved DAX Generation:** A major breakthrough was achieved by scoping child agents to specific tables and embedding schema instructions, allowing the AI to generate syntactically correct DAX queries from natural language.
- **Benchmark Results:** The prototype achieved a 100% success rate across 5 KPIs, including Security Compliance, Live Data Capability, and Query Accuracy.

4. Conclusion

This project successfully delivered two solutions that transform the planning team's workflow from reactive data validation to proactive strategic analysis.

For the primary deliverable, the Power BI report evolved into an automated violation tracker that visualizes 10 complex business rules. Significant technical challenges were overcome, specifically regarding the development of DAX logic for context-aware changeover calculations and two-step duplicate order validation. The integration of a custom Power App for logging "Agreed Violations" further transformed the report from a passive dashboard into an active workflow tool, preventing redundant discussions in meetings.

For the secondary deliverable, the feasibility study provided a data-driven recommendation to adopt Microsoft Copilot Studio, as it was the only platform capable of meeting enterprise requirements for security, in-tenant data processing, and live connectivity. This recommendation was validated through a high-fidelity prototype which achieved a 100% success rate across all benchmark KPIs, including query accuracy and live data retrieval.

The combination of these tools has laid the foundation for a more efficient planning process, shifting the focus of weekly meetings from verifying data to making strategic decisions.